

17 Carbonyl compounds

17.1 Aldehydes and ketones

Learning outcomes

Candidates should be able to:

- recall the reactions (reagents and conditions) by which aldehydes and ketones can be produced:
 - the oxidation of primary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce aldehydes
 - the oxidation of secondary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce ketones
- describe:
 - the reduction of aldehydes and ketones using NaBH_4 or LiAlH_4 to produce alcohols
 - the reaction of aldehydes and ketones with HCN , KCN as catalyst, and heat to produce hydroxynitriles as exemplified by ethanal and propanone
- describe the mechanism of the nucleophilic addition reactions of hydrogen cyanide with aldehydes and ketones in 17.1.2(b)
- describe the use of 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) to detect the presence of carbonyl compounds
- deduce the nature (aldehyde or ketone) of an unknown carbonyl compound from the results of simple tests (Fehling's and Tollens' reagents; ease of oxidation)
- deduce the presence of a $\text{CH}_3\text{CO}-$ group in an aldehyde or ketone, $\text{CH}_3\text{CO}-\text{R}$, from its reaction with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate of tri-iodomethane and an ion, RCO_2^-

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